

# Viscosity Curves in Plastic Injection Molding

*A Technical Training Manual for Engineers, Process Technicians, Tooling Teams, and Manufacturing Leaders*

**Version 1.0 - Prepared for shop-floor training and technical onboarding**

Important limitation: this document is source-backed and technically reviewed against cited references, but it has not been externally peer-reviewed by SPE, RJG, Beaumont, Autodesk, Moldex3D, resin suppliers, or machine OEMs.

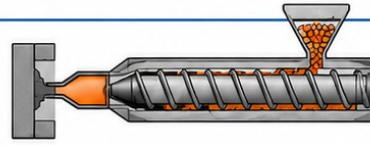
Use plant procedures, OEM manuals, supplier datasheets, and qualified engineering sign-off before applying changes to production molds.

# Viscosity Curves in Injection Molding

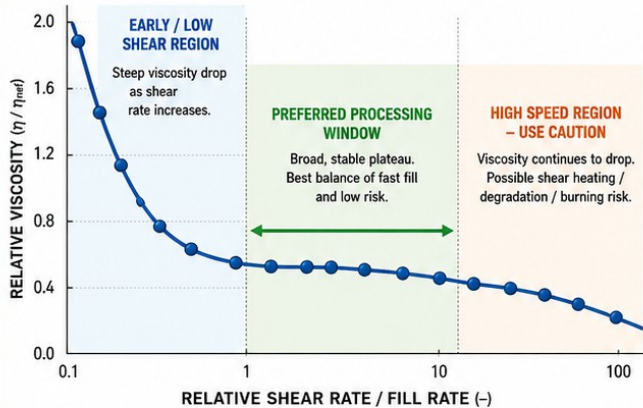
## Definition, Purpose, and Interpretation

### 1 DEFINITION

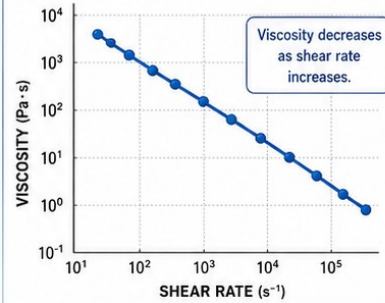
A viscosity curve in scientific injection molding is a plot used to show how apparent (or relative) viscosity changes as plastic flow rate or shear rate changes during controlled short-shot tests. Polymers are typically shear-thinning—meaning higher shear rate usually lowers apparent viscosity.



### 2 RELATIVE VISCOSITY vs RELATIVE SHEAR RATE / FILL RATE



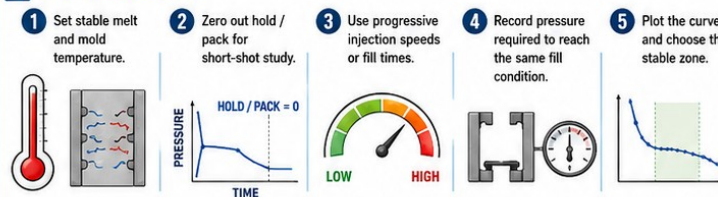
### 3 SHEAR RATE vs VISCOSITY (TYPICAL POLYMER)



#### TERMS

- **Relative Viscosity ( $\eta / \eta_{ref}$ )**: Viscosity normalized to a reference point (chosen in the stable zone).
- **Relative Shear Rate / Fill Rate (-)**: Shear rate at a given speed normalized to the reference speed.
- **Short-Shot Test**: Shots are stopped at the same fill condition (e.g., 95% fill) while changing injection speed.

### 4 HOW THE DOE IS RUN



### 5 WHY RUN A VISCOSITY CURVE?

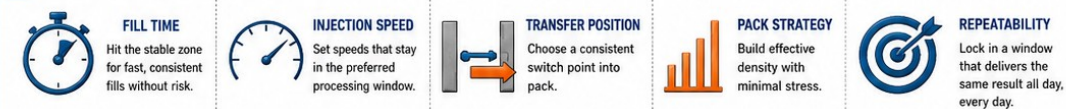
- ✓ Establish a robust fill speed window.
- ✓ Reveal viscosity sensitivity to speed.
- ✓ Reduce lot-to-lot surprises.
- ✓ Support process transfer.
- ✓ Improve consistency in hot and cold runner molds.

**KEY RULE: ONLY ONE MAJOR VARIABLE SHOULD CHANGE DURING THE STUDY.**  
Keep the following stable throughout the test:

- ✓ Resin (material grade & lot)
- ✓ Cushion / screw position
- ✓ Melt temperature
- ✓ Part fill target (same short-shot length)
- ✓ Mold temperature
- ✓ Machine condition (screw, check ring, wear)



### 6 THE OUTCOME HELPS YOU OPTIMIZE



Use viscosity curves as a proven scientific tool to understand your material and establish a robust, repeatable process.

Training infographic: definition, method, and interpretation of an in-mold viscosity curve.

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# 1. Executive Summary and Learning Objectives

A viscosity curve, also called an in-mold rheology curve or relative viscosity curve, is one of the foundational studies in scientific injection molding. It is used to select a robust fill speed or fill time by observing how the resin-mold-machine system responds as injection velocity changes. The study is not merely a material test. It is a system test: resin, dryer, barrel, screw, check ring, nozzle, hot runner or cold runner, gate, venting, wall thickness, clamp, cooling, and data collection discipline all affect the result.

The most important training point is this: a viscosity curve does not produce a magic speed setting. It produces a controlled understanding of the stable filling window. A good processor uses the curve to select a fill speed that is fast enough to avoid unstable low-shear filling, but not so fast that shear heating, pressure spikes, flash, jetting, burning, material damage, or machine limitations dominate the process.

- Define apparent viscosity, relative viscosity, shear rate, fill rate, shear thinning, and process window in plain shop-floor language and technical language.
- Perform a repeatable viscosity curve DOE without contaminating the data by changing multiple variables.
- Distinguish cold-runner pressure losses from hot-runner thermal and residence-time effects.
- Interpret the curve to choose a fill speed/fill time and then build transfer, pack/hold, gate seal, cooling, and process window studies around that foundation.
- Recognize false curves caused by moisture, drifting melt temperature, bad check rings, poor cushion control, unstable hot-runner zones, inadequate clamp force, and inconsistent short-shot targets.

## 1.1 The processor mindset

The curve is a decision-support tool. It tells the processor how sensitive the process is to flow rate under controlled conditions. It does not replace observation, part inspection, cavity pressure data, mold-filling simulation, or engineering judgment. It also does not excuse unsafe operation. A curve point that flashes the mold, over-pressures the system, burns material, damages a gate, or drives the press into an uncontrolled pressure limit is not a valid process point; it is a warning.

# 2. Technical Definition and Core Theory

## 2.1 Precise technical definition

In plastic injection molding, a viscosity curve is a plot of apparent or relative melt viscosity against shear rate, relative shear rate, injection fill rate, or fill time, generated by molding controlled short shots at different injection velocity settings while holding the material, temperature, shot target, transfer condition, cushion, and machine condition as constant as practical. In scientific molding, the in-mold version typically uses actual fill time and injection pressure at transfer or at the short-shot endpoint as proxy variables for shear rate and viscosity. The result is used to identify a region where changes in speed produce minimal change in apparent viscosity and where part fill is stable and repeatable.

External laboratory rheology curves are generated on capillary or slit-die rheometers and report apparent or corrected viscosity as a function of shear rate and temperature. In-mold rheology curves use the production mold and press, so they include real pressure losses and non-isothermal cooling/frozen-layer effects. Those two curves are related, but they are not interchangeable.

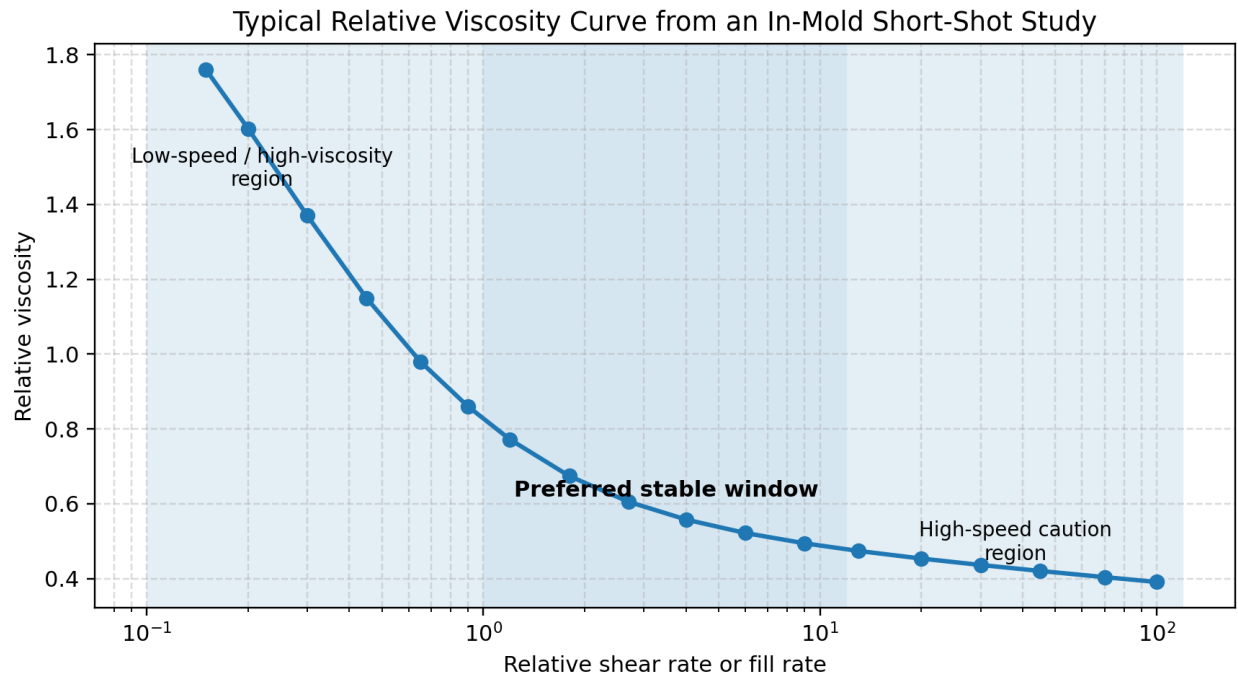


Figure 1. Typical relative viscosity curve. The preferred window is the flatter region, not necessarily the fastest possible injection speed.

## 2.2 Apparent viscosity versus true material viscosity

True viscosity is a material property measured under defined temperature, pressure, shear rate, and geometry. Apparent viscosity in a production mold is a process response. It includes resin rheology plus losses through the machine nozzle, sprue bushing, cold runner, hot runner manifold, hot drops, tip geometry, gates, vents, thin walls, changes in frozen-layer thickness, and sometimes machine-control limitations. This is why a viscosity curve is best described as a resin-mold-machine viscosity response curve.

## 2.3 Shear thinning and non-Newtonian behavior

Most thermoplastic melts used in injection molding are non-Newtonian and shear-thinning; apparent viscosity decreases as shear rate increases. At low shear rates, polymer chains are more entangled and resist flow. As shear rate increases, chains orient and disentangle in the direction of flow, which reduces resistance. At very high rates, extra shear heating, melt fracture, material degradation, gate blush, jetting, or burning may occur depending on material and tool geometry.

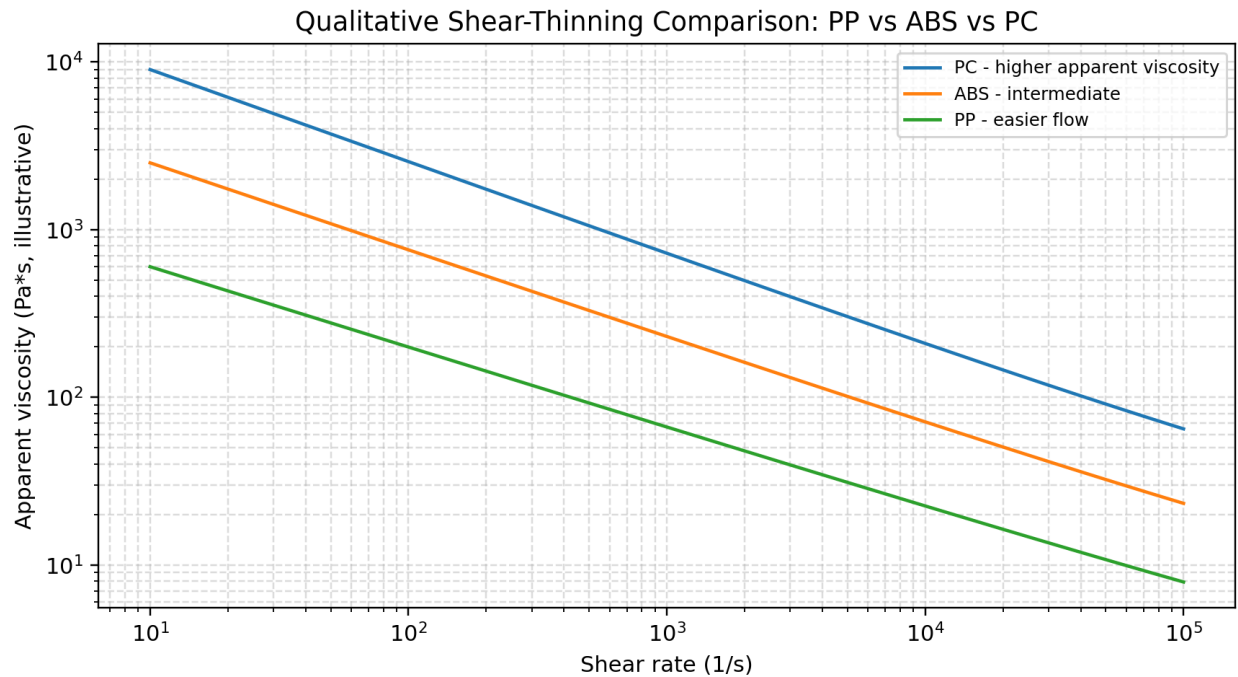


Figure 2. Qualitative material-family comparison. Actual curves must come from the resin grade, moisture state, temperature, and tool being processed.

## 2.4 Common in-mold calculations

Exact shear rate in a mold cavity is difficult to calculate because the melt front, frozen layer, wall thickness, gate geometry, and local flow length change during filling. Therefore many scientific-molding workflows use relative values. The most common shop-floor approach is:

- Relative shear rate or relative fill rate: proportional to  $1 / \text{fill time}$ , or normalized fill rate against a reference point.
- Relative viscosity: proportional to plastic pressure at transfer multiplied by fill time. If the machine reports hydraulic pressure, multiply by the intensification ratio to estimate plastic pressure.
- Relative viscosity index:  $RV = P_{\text{plastic}} \times \text{fill time}$ . If hydraulic pressure is used:  $RV = P_{\text{hydraulic}} \times \text{intensification ratio} \times \text{fill time}$ .

Important: compare points only when the short-shot target is the same. If one point is 80% full and another is 96% full, pressure is not comparable because the melt has traveled different distances, seen different cooling, and encountered different cavity restrictions.

## 2.5 Relationship to fill time

Fill time is the practical control variable many processors keep with the mold. Two presses may require different velocity setpoints to produce the same fill time because of machine response, screw diameter, hydraulic/electric drive behavior, material compressibility, intensification ratio, and check-ring condition. For process transfer, the target should be plastic variables: fill time, pressure response, transfer position, part weight, cavity pressure if available, and melt/mold temperature - not merely the percent injection speed displayed by one machine.

### 3. Why Viscosity Curves Are Performed

The root cause for running a viscosity curve is uncertainty. The processor needs to know how the actual system reacts when flow rate changes. Without this study, the chosen speed may sit on a steep part of the curve where a small viscosity shift creates large pressure, fill, dimension, weld strength, gloss, or flash changes. The curve is a controlled way to expose that sensitivity before a production process is locked.

- Select a robust first-stage fill speed or fill time.
- Find the region where apparent viscosity changes the least with speed.
- Reduce sensitivity to lot-to-lot material variation.
- Establish a transferable process using plastic variables rather than arbitrary machine settings.
- Avoid low-speed hesitation, unstable weld lines, cold slugs, excessive pressure, or incomplete filling.
- Avoid high-speed shear heating, gate blush, burning, jetting, flash, over-clamping demand, and material degradation.
- Document a defensible process baseline before pack/hold, gate seal, cooling, cavity balance, or full process-window studies.

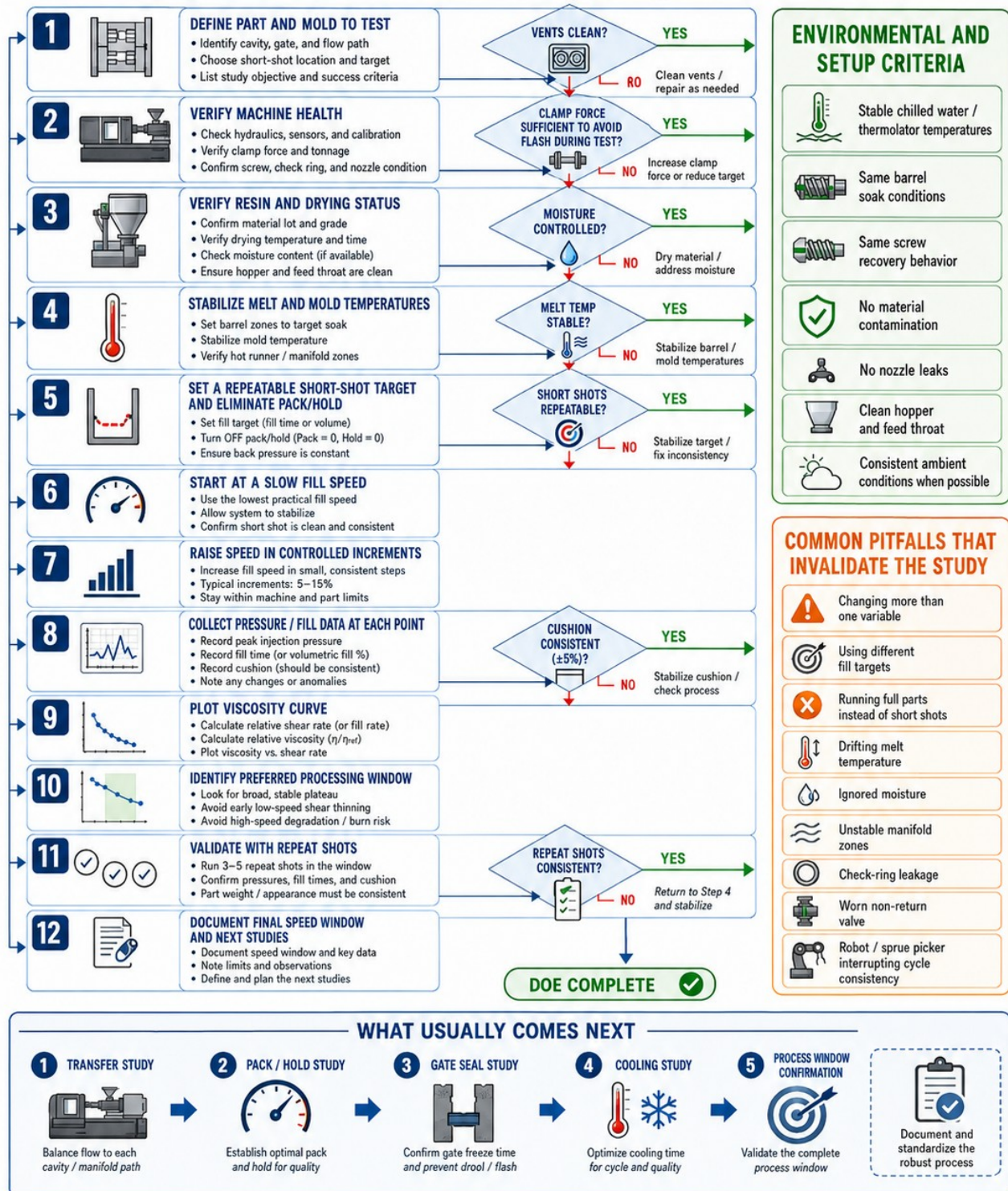
### 4. The Correct DOE Method

A viscosity curve is a one-factor DOE. The factor intentionally changed is injection speed or target fill time. Everything else must be held constant or documented. If the process team changes melt temperature, hot-runner setpoints, back pressure, decompression, transfer target, pack pressure, mold cooling, material lot, regrind level, or short-shot fill percentage during the study, the curve becomes contaminated.



# Viscosity Curves in Injection Molding

Step-by-Step DOE Flowchart — Before, During, and After the Study



Follow this flow, control one variable at a time, and let the data guide your decisions.

Figure 3. Viscosity study SOP flowchart. Use as a training poster and pre-run checklist.



## 4.1 Preconditions before starting

- Use the correct resin grade, color, additive package, and regrind policy. Record lot number and moisture condition.
- Dry hygroscopic materials to resin supplier requirements. For PC, PA, PET, PBT, TPU, and many engineered materials, moisture can destroy the validity of the study.
- Verify that the machine is not pressure-limited at the study points. A pressure-limited shot does not represent the commanded fill speed.
- Verify check ring/non-return valve consistency using cushion repeatability and pressure response. A leaking check ring creates false low pressure and inconsistent fill.
- Stabilize barrel temperatures, actual melt temperature, mold temperature, and hot-runner setpoints. Do not rely only on controller setpoints.
- Use a clean barrel, clean hopper, clean feed throat, correct screw recovery, stable back pressure, and consistent decompression.
- Confirm clamp force is adequate. Flashing during the study corrupts pressure data and can damage the tool.
- Confirm vents are clean. Poor venting increases pressure artificially and can make a good speed look bad.
- Confirm short-shot target and safety limits. Many processors use roughly 90-95% fill with pack/hold off, but plant procedure and part risk must govern.

## 4.2 Recommended sequence

1. Establish safe machine limits: maximum injection pressure, maximum velocity, clamp tonnage, part/tool risk, and material degradation limits.
2. Set pack/hold to zero or disable pack/hold so the study measures fill behavior, not packing behavior. Use adequate cooling time for safe ejection.
3. Set transfer to a consistent position or short-shot endpoint that produces the same fill percentage at each speed. The endpoint is the comparison anchor.
4. Start at a safe low-to-moderate speed, then move in controlled increments. Some plants prefer high-to-low to reduce risk of sticking at very slow fill; follow local safe SOP.
5. At each point, allow the process to stabilize and record at least three shots if the tool and schedule permit. Record peak injection pressure, pressure at transfer, fill time, cushion, screw recovery, melt temperature, mold temperature, and part weight if applicable.
6. Plot relative viscosity versus relative shear rate/fill rate. Identify the broadest stable region where viscosity changes least with speed and part appearance remains acceptable.
7. Select a fill speed inside the stable region, not on the edge. Repeat validation shots at that setting.
8. Then run follow-up studies: transfer position, pack/hold, gate seal, cooling, cavity balance, process window, and dimensional capability.

## 4.3 Data that must be recorded

| Record                                                              | Why it matters                                                  | What goes wrong if ignored                           |
|---------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|
| Machine ID, screw diameter, intensification ratio                   | Needed for process transfer and pressure conversion.            | Cannot compare data across presses.                  |
| Material grade, lot, color, regrind %, dryer conditions             | Material changes shift the curve.                               | False conclusion that speed is the root cause.       |
| Actual melt temperature and barrel setpoints                        | Temperature shifts viscosity.                                   | Curve moves without speed being responsible.         |
| Mold temperature: supply, return, flow, actual steel where possible | Skin freeze and cavity resistance change with mold temperature. | Pressure readings become geometry/thermal artifacts. |
| Hot-runner manifold/tip zones                                       | Hot runner is part of the flow path.                            | Wandering zones create erratic points.               |

|                                                  |                                                                        |                                                                          |
|--------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Injection speed command and actual fill time     | Speed command alone may not equal actual flow rate.                    | Cannot transfer process to another press.                                |
| Peak injection pressure and pressure at transfer | Required to calculate relative viscosity and detect pressure limiting. | Invalid data if pressure is capped.                                      |
| Transfer position and short-shot target          | Comparison requires identical fill condition.                          | Different flow lengths produce non-comparable data.                      |
| Cushion and screw recovery time                  | Detects check-ring and shot-size consistency.                          | Leaky check ring or inconsistent recovery hides true viscosity.          |
| Part observations and defects                    | The window must make good parts, not just a pretty graph.              | Speed may be selected in a region that burns, flashes, or damages parts. |

## 5. Calculations, Data Sheet, and Acceptance Criteria

The goal is not mathematical elegance; the goal is repeatable decision-making. The calculations should be simple enough that a technician can run them at the press, but disciplined enough that engineering can audit them later.

### 5.1 Example calculation

| Point | Injection speed setting | Fill time (s) | Plastic pressure at endpoint (psi) | Relative fill rate (1/t normalized) | Relative viscosity normalized |
|-------|-------------------------|---------------|------------------------------------|-------------------------------------|-------------------------------|
| A     | 25                      | 1.20          | 12,000                             | 1.00                                | 1.00                          |
| B     | 35                      | 0.95          | 10,000                             | 1.26                                | 0.79                          |
| C     | 50                      | 0.76          | 8,800                              | 1.58                                | 0.56                          |
| D     | 70                      | 0.62          | 8,400                              | 1.94                                | 0.43                          |
| E     | 90                      | 0.54          | 8,900                              | 2.22                                | 0.40                          |

In this illustrative data set, the lower-speed points show a large reduction in apparent viscosity as speed increases. Points C-D are flatter and are likely candidates for the preferred speed window. Point E may still look acceptable, but if defects or pressure instability appear, the processor should not select it just because it is faster.

## 6. Material Factors: PP, ABS, PC, Fillers, Additives, Regrind, and Moisture

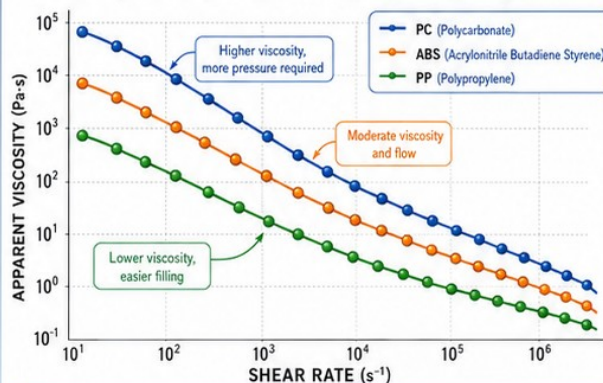
# Viscosity Curves in Injection Molding

## Polymer Behavior Comparison — PP vs ABS vs PC



### 1 APPARENT VISCOSITY vs SHEAR RATE

All three polymers are shear-thinning—viscosity decreases as shear rate increases. PC has the highest viscosity, ABS is intermediate, and PP flows most easily.

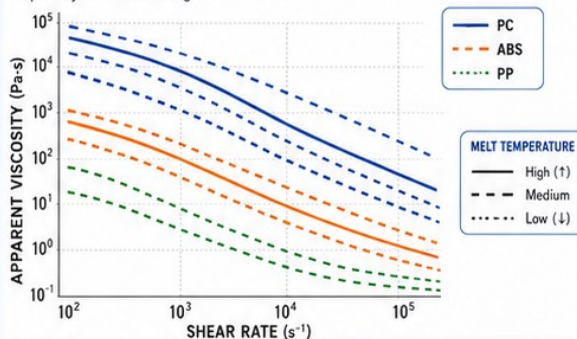


#### TERMS

- **Apparent Viscosity ( $\eta_{app}$ ):** Measured viscosity of the melt at a given shear rate and temperature.
- **Shear-Thinning:** Viscosity decreases as polymer chains align with flow.

### 2 TEMPERATURE SENSITIVITY

Increasing melt temperature reduces viscosity for all polymers. PC requires higher melt temperature and is more pressure-sensitive, especially in thin-wall filling.



#### PP (Polypropylene)

- ✓ Broad processing window
- ✓ Less pressure-sensitive
- ✓ Easier to fill thin sections

#### PC (Polycarbonate)

- ✓ Narrow processing window
- ✓ Higher pressure-sensitive
- ✓ Needs higher melt temperature; watch thin-wall filling

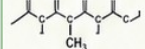
### 3 MATERIAL COMPARISON — HOW POLYMER FAMILY CHANGES VISCOSITY CURVE BEHAVIOR

| Material                                                     | Typical flow behavior                                                             | Sensitivity to shear                                                 | Sensitivity to moisture                                    | Hot-runner considerations                                                         | Cold-runner considerations                                             | Common risks during the study                                                                                                                                                                                           |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PP</b><br>(Polypropylene)<br>Semi-crystalline             | Low viscosity; easy flow.<br>Strong shear-thinning.<br>Longer relaxation.         | Low-Moderate<br>Viscosity drops quickly then flattens at high shear. | Low<br>Not hygroscopic.<br>Minimal drying requirements.    | Lower temperatures.<br>Less residence time sensitivity. Minimal degradation risk. | Low pressure drop.<br>Short flow paths typically fine.                 | <ul style="list-style-type: none"><li>• Over-speeding can cause flashing.</li><li>• Melt fracture at very high shear.</li><li>• Inconsistent pack due to low melt strength.</li></ul>                                   |
| <b>ABS</b><br>(Acrylonitrile Butadiene Styrene)<br>Amorphous | Moderate viscosity.<br>Balanced flow and toughness.<br>Good shear-thinning.       | Moderate<br>Viscosity decreases steadily with shear.                 | Moderate<br>Absorbs some moisture; dry if molded wet.      | Use moderate temps.<br>Watch for residence time and thermal oxidation.            | Moderate pressure drop. Balance runner size and material temp.         | <ul style="list-style-type: none"><li>• Brittle appearance if molded too cold.</li><li>• Jetting / splay with high speed.</li><li>• Discoloration from degradation.</li></ul>                                           |
| <b>PC</b><br>(Polycarbonate)<br>Amorphous                    | High viscosity.<br>Strong shear-thinning but remains high.<br>Shorter relaxation. | High<br>Very pressure-sensitive, especially in thin walls.           | High<br>Hygroscopic.<br>Requires drying (see caution box). | Higher temperatures needed. Control residence time carefully.                     | High pressure drop. Runner heat loss increases viscosity and pressure. | <ul style="list-style-type: none"><li>• High pressure / clamp tonnage.</li><li>• Short shots in thin-wall features.</li><li>• Splay, weld lines from pressure spikes.</li><li>• Silver streaks from moisture.</li></ul> |

### 4 KNOW YOUR MATERIAL

#### PP (Polypropylene)

Semi-crystalline



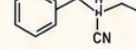
Lower density • Floats in water

Common in: living hinges, caps, containers, closures

Fast cooling, crystallization influences shrinkage & fill.

#### ABS (Acrylonitrile Butadiene Styrene)

Amorphous



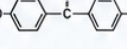
Balanced toughness & flow

Common in: appliances, housings, electronics, toys

No crystallization; cooling is smoother and predictable.

#### PC (Polycarbonate)

Amorphous



High toughness & heat resistance

Common in: lenses, guards, medical devices, safety parts

High processing temp; flows high viscosity, pressure-sensitive.

Semi-crystalline vs. amorphous behavior changes how flow and cooling are observed and interpreted.

### 5 ADDITIVES & FILLERS — EFFECT ON VISCOSITY

| Additive / Filler                         | Typical Effect on Apparent Viscosity | Effect on Shear Response                                     | Notes                                                 |
|-------------------------------------------|--------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|
| Glass Fiber (10–40%)                      | Increase ↑↑                          | Can increase shear-thinning (more slope) at medium shear.    | Raises viscosity & pressure; improves stiffness.      |
| Mineral Filler (CaCO <sub>3</sub> , Talc) | Increase ↑                           | Slight change or increased shear-thinning.                   | Adds stiffness; can raise melt temp.                  |
| Color Concentrate (Pigment)               | Slight Increase ↑                    | Small effect                                                 | Check carrier compatibility.                          |
| Lubricant / Flow Promoter                 | Decrease ↓                           | Reduces viscosity across shear range.                        | Improves flow, reduces pressure & wear.               |
| Regrind                                   | Variable ↓                           | Can increase or decrease depending on history & degradation. | Melt history matters—watch moisture & thermal damage. |

↑↑ Strong Increase    ↑ Increase    ↓ Decrease    ↓ Variable



#### MOISTURE CAUTION

Moisture-sensitive materials (especially PC) require drying discipline.

- Hydrolytic degradation raises viscosity.
- Splay, silver streaks, and poor appearance.
- Inconsistent viscosity corrupts the study.
- Follow resin supplier drying guidelines.



### 6 PRACTICAL TAKEAWAY

- ✓ **PP:** Wider stable speed window. Run higher speeds to move through the window into the flat region.
- ✓ **ABS:** Moderate window. Use mid speeds; watch for pressure rise at the high end.
- ✓ **PC:** Narrow window. Use lower/mid speeds; pressure escalates quickly at high shear.
- ✓ Always re-check temperature, pressure, and fill quality for each material family.
- ✓ Don't compare curves across materials at the same pressure—compare shape and window.



Different polymer families behave differently. Understand the curve, respect the window, and optimize with confidence.

Figure 4. Polymer family comparison for training. Use grade-specific data for production decisions.



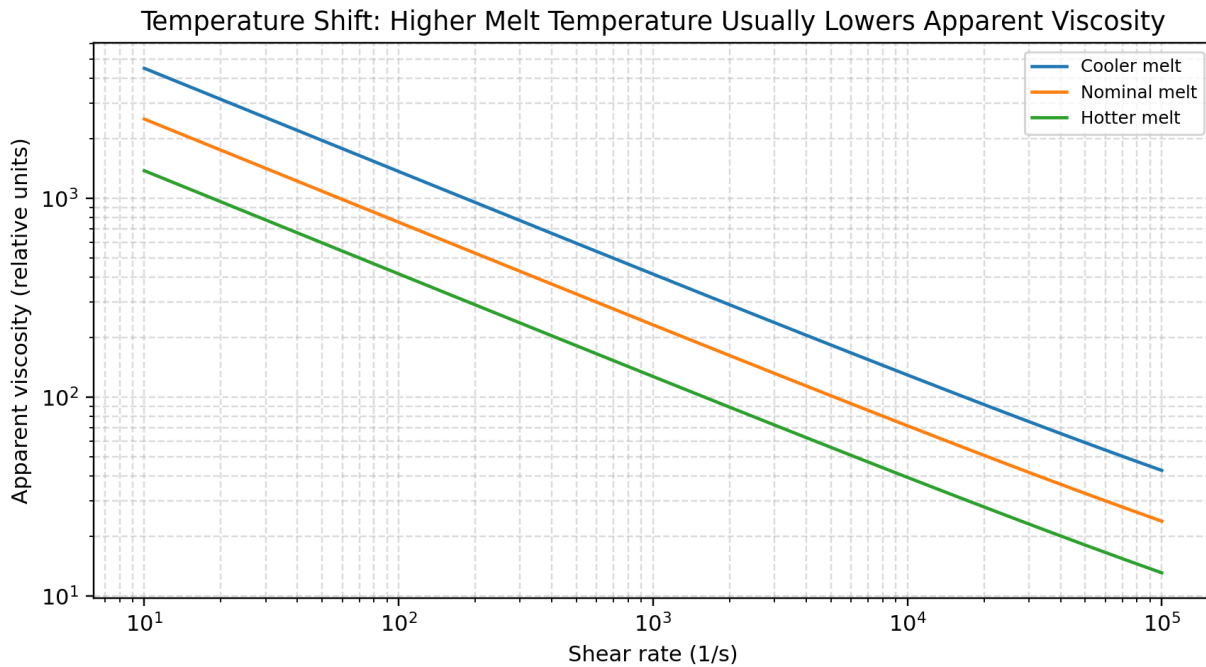


Figure 5. Temperature effect. Increasing melt temperature generally lowers viscosity, but excessive heat can degrade material or alter crystallization/cooling behavior.

## 6.1 Resin family and molecular architecture

Polymer family determines baseline viscosity, shear-thinning behavior, temperature sensitivity, moisture sensitivity, crystallization, and degradation risk. Within the same family, grade selection can be more important than the family label: a high-flow PC may process differently than a low-flow ABS; a filled PP can require more pressure than an unfilled engineering resin in a given tool.

| Material           | Typical viscosity behavior                                                                          | Common uses                                                | Cold-runner implications                                                                                                          | Hot-runner implications                                                                                 | Study risks                                                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| PP - Polypropylene | Semi-crystalline; usually relatively easy flow; often strong shear-thinning; sets up quickly.       | Closures, caps, living hinges, containers, consumer goods. | Cold runners are often feasible because pressure drop is manageable, but runner cooling and gate freeze can strongly affect fill. | Hot runner temperatures are usually lower than PC/ABS; avoid drool, stringing, and overheating at tips. | Flash at high speed, gate blush, inconsistent packing due to low melt strength, shrink/warp sensitivity. |
| ABS                | Amorphous; moderate viscosity; good toughness/flow balance; sensitive to shear and thermal history. | Housings, appliances, automotive trim, electronics, toys.  | Runner balance and heat loss matter; moderate pressure drop; watch weld lines and surface appearance.                             | Watch residence time, thermal oxidation, color shift, and tip balance.                                  | Splay, discoloration, jetting, gloss issues, brittle appearance if too cold.                             |
| PC - Polycarbonate | Amorphous; high viscosity; needs higher                                                             | Lenses, guards, medical parts, safety parts,               | Cold runners add major pressure and                                                                                               | Hot runners must control residence time                                                                 | Short shots, high clamp demand, silver streaks                                                           |

|  |                                                                             |           |                                                      |                                                    |                                                 |
|--|-----------------------------------------------------------------------------|-----------|------------------------------------------------------|----------------------------------------------------|-------------------------------------------------|
|  | melt temperature; pressure-sensitive especially in thin walls; hygroscopic. | housings. | heat-loss burden; large runners/gates may be needed. | and avoid stagnant/degraded melt at tips/manifold. | from moisture, burn marks, degradation, stress. |
|--|-----------------------------------------------------------------------------|-----------|------------------------------------------------------|----------------------------------------------------|-------------------------------------------------|

## 6.2 Molecular weight and melt flow rate

Higher molecular weight generally increases viscosity and reduces melt flow rate. Lower MFR/MFI materials usually need more pressure or higher temperature/speed. Do not compare MFR alone as a full rheology curve; MFR is measured at one load and temperature, while injection molding spans high shear rates and non-isothermal flow.

## 6.3 Additives and lubricants

Internal lubricants and flow promoters can lower apparent viscosity, but they may affect weld strength, plating, painting, migration, screw slip, gate vestige, or long-term properties. Validate through testing, not assumptions.

## 6.4 Fillers and reinforcements

Glass fiber, mineral filler, talc, calcium carbonate, flame retardants, impact modifiers, and pigments can increase viscosity, alter shear-thinning slope, increase pressure demand, change thermal conductivity, and increase wear. Fiber orientation also affects shrink and warpage.

## 6.5 Regrind

Regrind can shift viscosity upward or downward depending on thermal history, moisture, chain scission, contamination, fines, and color/additive history. For the viscosity curve, either eliminate regrind or hold the percentage, particle size, drying, and blend method constant.

## 6.6 Moisture

Moisture can create bubbles, splay, silver streaks, hydrolysis, molecular-weight reduction, and viscosity changes. Hygroscopic materials such as PC and PA require drying discipline. A wet material can produce a curve that appears to be a speed problem while the root cause is material condition.

## 6.7 Lot-to-lot variation

A properly selected speed window reduces sensitivity to resin variation, but it does not eliminate it. If new lots shift pressure or fill time, compare actual melt temperature, moisture, MFR/rheology data, dryer performance, screw recovery, and pressure trace before adjusting the process.



## 7. Process Parameters and Their Influence

# Viscosity Curves in Injection Molding

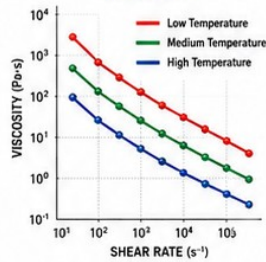
What Shifts the Curve? Material, Process, Mold, Machine, and Auxiliary Factors

### 1 MATERIAL FACTORS

- Resin family**  
Polymer chemistry and architecture drive flow behavior.
- Molecular weight**  
Higher MW increases viscosity and reduces flow.
- Melt flow rate (MFR)**  
Lower MFR resins are more viscous.
- Additives**  
Lubricants lower viscosity. Impact modifiers can raise it.
- Fillers**  
Fillers increase viscosity and reduce flow.
- Regrind content**  
Regrind raises viscosity and narrows the processing window.
- Moisture sensitivity**  
Moisture can cause hydrolysis, bubbles, and higher viscosity.
- Lot-to-lot variation**  
Different lots can shift the whole curve.
- Thermal history**  
Overheating or long residence time can degrade and change viscosity.

PP is usually more shear-thinning and flows easier than PC at comparable conditions.  
ABS is intermediate and sensitive to temperature and shear history.

#### TYPICAL EFFECT OF TEMPERATURE ON VISCOSITY



### MATERIAL COMPARISON (Typical)

| MATERIAL                                        | RELATIVE FLOW TENDENCY (at same conditions) | TEMP. SENSITIVITY of VISCOSITY | VISCOSITY TREND with SHEAR RATE   |
|-------------------------------------------------|---------------------------------------------|--------------------------------|-----------------------------------|
| <b>PP</b><br>(Polypropylene)                    | High<br>(flows easiest)                     | Moderate                       | Strong shear-thinning             |
| <b>ABS</b><br>(Acrylonitrile Butadiene Styrene) | Medium<br>(intermediate)                    | High                           | Moderate to strong shear-thinning |
| <b>PC</b><br>(Polycarbonate)                    | Low<br>(more viscous)                       | Moderate to High               | Moderate shear-thinning           |

### 2 PROCESS PARAMETERS

- Injection speed / fill time**  
Changes shear rate. Higher speed = higher shear.
- Peak pressure**  
Affects packing load and can influence melt compression.
- Melt temperature**  
Shifts the entire viscosity curve down (hotter) or up (cooler).
- Mold temperature**  
Affects skin freeze rate and apparent viscosity.
- Cushion**  
Changes pack/hold volume and pressure conditions.
- Transfer position**  
Controls where packing begins; affects pressure build and recovery.
- Cooling time**  
Affects part temperature at ejection and next shot recovery.
- Decompression (stroke)**  
Relieves pressure, prevents drool, and stabilizes next shot.

Changing speed changes shear rate. Changing melt temperature shifts the whole curve downward (hotter) or upward (cooler).

### 3 MOLD DESIGN FACTORS

- Gate size**  
Smaller gates increase shear and pressure.
- Gate location**  
Unbalanced location changes flow path and shear.
- Runner balance**  
Unbalanced runners create uneven shear and pressure.
- Wall thickness variation**  
Thin sections increase shear and distort results.
- Venting**  
Adequate vents prevent gas compression and false data.
- Air traps**  
Air compression raises required pressure and masks flow.

Restrictive gates and thin sections raise required pressure and distort curve interpretation.

### 4 MACHINE FACTORS

- Screw design**  
Affects melting, mixing, shear, and residence time.
- Check ring condition**  
Worn or leaking check rings create pressure loss and false data.
- Barrel wear**  
Increases clearance and reduces pressure transmission.
- Nozzle condition**  
Worn or buildup nozzles change flow resistance and pressure.
- Back pressure**  
Higher back pressure increases shear and heat.
- Clamp tonnage**  
Insufficient clamp tonnage causes mold opening and data errors.

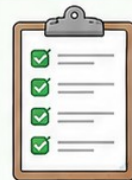
Worn screws, barrels, or leaking check rings create false data and shifting curves.

### 5 AUXILIARY EQUIPMENT

- Chiller**  
Keeps mold temperature stable.
- Thermolator**  
Maintains stable melt and mold temperature.
- Hot-runner controller**  
Controls zone temperatures and prevents drift.
- Robot / Sprue picker**  
Ensures consistent part removal time and cycle start conditions.
- Dryer**  
Removes moisture and prevents viscosity changes from humidity.

### WHAT A GOOD STUDY REQUIRES

- Stable resin (same lot, dried as required, minimal regrind).
- Stable temperature control (melt and mold).
- Repeatable shot-to-shot recovery (peak pressure and cushion).
- Identical short-shot target for all points.
- Minimal outside disturbances (people, environment, power).



### FALSE-SHIFT WARNING – Common Causes of a Misleading Curve

- Moisture**  
Increase in moisture raises viscosity and causes bubbles.
- Drifting melt temperature**  
Hotter or cooler melt shifts the entire curve.
- Inconsistent cushion**  
Changing cushion changes pressure and recovery.
- Changing fill target**  
Different short-shot lengths create non-comparable points.
- Sticky or leaking check ring**  
Pressure loss alters measured pressure and viscosity.
- Unstable hot-runner zone**  
Temperature drift changes viscosity and flow.
- Poor venting or air traps**  
Gas compression raises pressure and masks flow.

Control the variables. Measure cleanly. Interpret correctly. Repeatability is the foundation of a reliable viscosity curve.

Figure 6. Influence groups that shift curve results. Use this as a setup checklist before the DOE.

Parameter

Primary effect

Training caution

|                                 |                                                                                                             |                                                                                                      |
|---------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Injection speed / fill time     | Primary DOE variable. Raises shear rate and usually lowers apparent viscosity until process risks dominate. | Changing speed also changes shear heating, orientation, pressure response, and cosmetic behavior.    |
| Injection pressure limit        | Should be high enough that velocity control is maintained during the study.                                 | A pressure-limited point is invalid because commanded speed was not achieved.                        |
| Melt temperature                | Higher melt temperature generally lowers viscosity and shifts the curve downward.                           | Too high can degrade material; too low can cause short shots, hesitation, stress, and weld weakness. |
| Mold temperature                | Controls frozen-layer development and surface replication.                                                  | Cold mold can make a resin look more viscous; hot mold can reduce pressure but extend cycle.         |
| Back pressure                   | Affects melt mixing, temperature, screw recovery, and air removal.                                          | Changing it during the study changes material condition and can shift results.                       |
| Screw recovery and cooling time | Affects residence time, melt homogeneity, and shot-to-shot readiness.                                       | Inconsistent recovery creates pressure and cushion noise.                                            |
| Transfer position               | Defines where fill ends and pack starts.                                                                    | For viscosity curve, transfer/endpoint must be consistent.                                           |
| Pack/hold pressure              | Should be off/zero for the fill study.                                                                      | Pack masks fill behavior and changes part weight, gate seal, pressure, and stress.                   |
| Decompression/suckback          | Affects drool, stringing, next-shot consistency, and hot-runner gate behavior.                              | Changing decompression changes next shot melt state.                                                 |



## 8. Cold Runner vs Hot Runner Molds

# Viscosity Curves in Injection Molding

Cold Runner vs Hot Runner — Why the Difference Matters

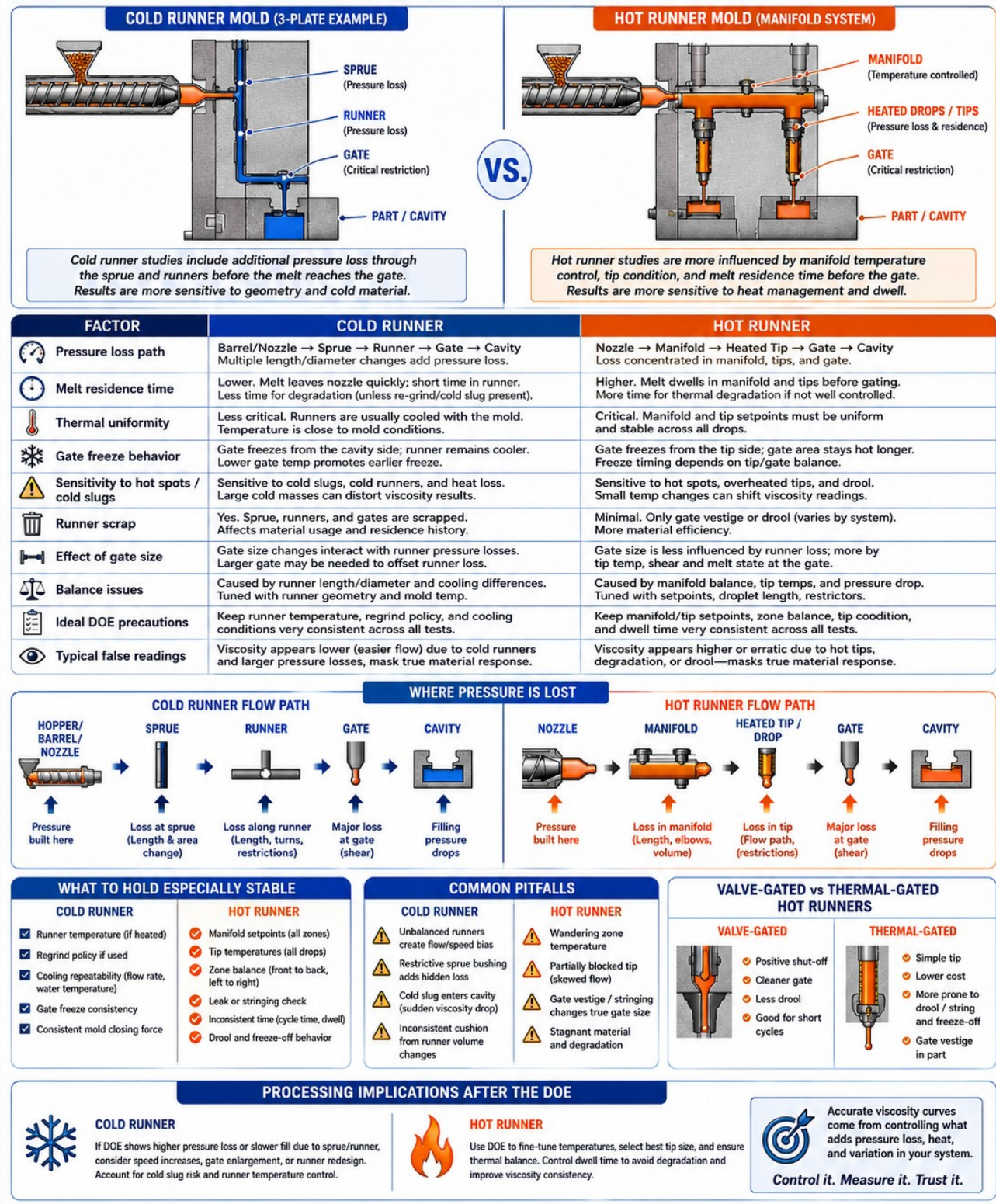


Figure 7. Cold-runner versus hot-runner implications during a viscosity curve study.

The cold-runner/hot-runner difference is not a detail; it changes how the curve must be interpreted. A cold runner adds cooled sprue and runner pressure losses before the melt reaches the gate. A hot runner removes most runner scrap but adds a heated manifold, heated drops, tips, possible valve pins, zone controls, and residence-time/degradation concerns. The same resin and cavity can produce different apparent viscosity response in cold-runner and hot-runner versions because the melt history before the gate is different.

| Factor             | Cold runner mold                                                                                                                             | Hot runner mold                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Main pressure path | Nozzle -> sprue -> runner -> gate -> cavity. Sprue and runners add length, turns, diameter changes, and heat loss.                           | Nozzle -> manifold -> drop/tip -> gate -> cavity. Losses concentrate in manifold, drops, tip geometry, valve-pin/tip restrictions, and gate. |
| Thermal risk       | Runner cools with mold; cold slug and frozen-layer effects can dominate.                                                                     | Hot manifold/tip temperatures dominate; stagnant melt and temperature imbalance can dominate.                                                |
| Residence time     | Usually lower in runner after injection, but total shot/residence still matters. Regrind from runner may return heat history to the process. | Higher because melt dwells in manifold and drops. Critical for heat-sensitive materials and color changes.                                   |
| Gate behavior      | Gate freezes from cavity/cold-runner side; gate size affects both fill pressure and freeze time.                                             | Thermal gate or valve gate behavior controls drool, stringing, vestige, freeze-off, and open/close timing.                                   |
| Balance            | Driven by runner length/diameter, gate dimensions, shear heating, and cooling differences.                                                   | Driven by manifold balance, heater zones, tip lengths, restrictors, valve timing, and material residence.                                    |
| False curve risks  | Cold slug, restrictive sprue bushing, unbalanced runner, inconsistent regrind, runner heat loss, gate freeze changes.                        | Wandering zone, blocked tip, leaking tip, degraded material, drool/stringing, valve timing, manifold pressure imbalance.                     |

## 8.1 Cold-runner viscosity-curve discipline

- Document sprue, runner, and gate geometry; they are part of the measured system response.
- Keep cooling water temperature and flow stable. Runner temperature affects pressure loss and cold slug behavior.
- Use consistent sprue/runner removal, robot/sprue picker timing, and cycle timing.
- Do not change regrind percentage during the curve. Runner regrind can affect viscosity and moisture.
- Watch for gate blush, hesitation, cold slugs, jetting from small gates, and pressure spikes from restrictive sprue bushings.

## 8.2 Hot-runner viscosity-curve discipline

- Stabilize all manifold and tip zones before collecting data. A drifting zone can mimic a material viscosity change.
- Confirm tips are not partially blocked and valve pins are functioning consistently.
- Record residence time and downtime. Heat soak can alter viscosity before the melt reaches the cavity.
- Watch for drool, stringing, gate freeze-off, gate vestige changes, and delayed valve opening.
- For multi-cavity hot runners, examine cavity balance; one bad drop can corrupt the curve if data is averaged across cavities.



## 9. Mold Design Factors

| Mold feature               | Effect on curve                                                                                             | Action                                                                      |
|----------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Gate size                  | Small gates raise shear and pressure; large gates reduce shear but may increase vestige and gate seal time. | Compare gate freeze, cosmetic requirements, and pressure capacity.          |
| Gate location              | Changes flow length, weld-line location, air trap location, pressure drop, and orientation.                 | A viscosity curve cannot fix a fundamentally bad gate location.             |
| Runner balance             | Unbalanced cold runners or hot manifolds cause cavities to see different fill rates.                        | Run cavity balance study after viscosity curve.                             |
| Wall thickness             | Thin areas increase shear and pressure; thick areas race-track and can hide hesitation elsewhere.           | Uniform wall thickness improves interpretability.                           |
| Venting                    | Poor vents compress air and increase pressure; burning may occur at high speed.                             | Clean and verify vents before the DOE.                                      |
| Surface finish and texture | Texture increases flow resistance and can reveal hesitation/gloss issues.                                   | Cosmetic parts may require a narrower speed window than the chart suggests. |
| Steel condition            | Worn gates, damaged vents, galling, residue, corrosion, and blocked water lines change the curve.           | Tool maintenance is part of process control.                                |

## 10. Machine Factors

| Machine factor                | Effect                                                                                                                         | Training action                                                               |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Screw design                  | Compression ratio, L/D, mixing section, barrier design, and material-specific design affect melt uniformity and shear history. | Use proper screw for resin family; record screw details for transfer.         |
| Screw/barrel wear             | Reduces melting efficiency and pressure transmission; increases recovery variation.                                            | Inspect when cushion and pressure vary without process explanation.           |
| Check ring / non-return valve | Leakage causes inconsistent cushion, shot volume, and pressure response.                                                       | Do not trust curve data from a bad check ring.                                |
| Nozzle                        | Buildup, mismatch, small orifice, cold slug, drool, or dead spots add pressure and degradation risk.                           | Clean and match nozzle/sprue; verify contact and temperature.                 |
| Clamp force                   | Insufficient clamp causes flash and pressure loss; excessive clamp can damage vents and parting line.                          | Verify clamp force and mold protection; do not solve flash only with tonnage. |
| Machine control response      | Hydraulic/electric response, pressure limit, velocity loop tuning, and screw diameter affect actual fill time.                 | Document actual fill time, not only speed percent.                            |

## 11. Auxiliary Equipment and Environmental Control

| Auxiliary system                 | Role in the curve                                                                     | Required control                                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Dryer                            | Controls moisture level and resin temperature.                                        | Monitor dew point, drying temperature, drying time, airflow, hopper size, and return-air filters. |
| Chiller / thermolator            | Controls mold and sometimes feed-throat/support temperatures.                         | Record supply/return temperature and flow; a setpoint alone is insufficient.                      |
| Hot-runner controller            | Controls manifold and tip zones.                                                      | Watch actual zone deviation, thermocouple placement, alarms, leaking heaters, and tuning.         |
| Robot / sprue picker             | Controls part removal timing and cycle consistency.                                   | Robot delays can change cooling time, resin residence, and startup consistency.                   |
| Conveyor / downstream automation | May affect cycle time if interlocked.                                                 | Any cycle interruption during the curve must be recorded.                                         |
| Material handling system         | Can introduce wrong resin, fines, dust, contamination, or inconsistent regrind blend. | Line clearance and labeling are process controls.                                                 |
| Plant environment                | Ambient temperature and humidity affect material moisture and mold cooling load.      | For critical validation, record ambient conditions.                                               |

## 12. Interpreting the Curve

# Viscosity Curves in Injection Molding

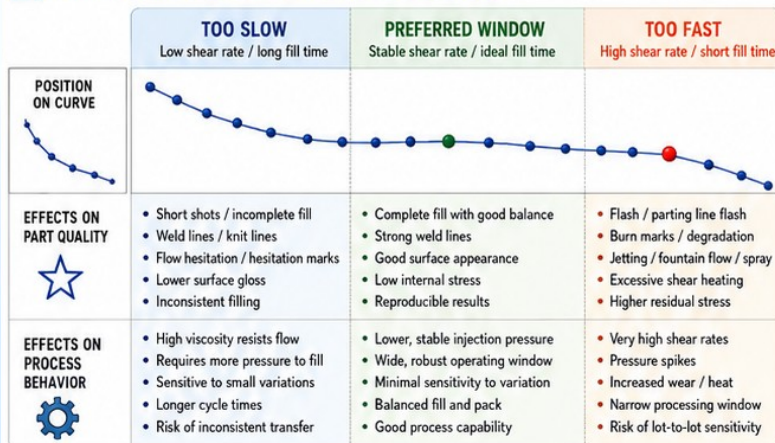
Using the Results — Optimization, Corrective Actions, and Preventive Measures

### 1 WHAT DOES THE VISCOSITY CURVE TELL US AND WHAT SHOULD WE DO WITH IT?

The viscosity curve shows how the resin flows at different shear rates (driven by injection speed). Use it to choose a robust speed window that fills the part well without creating defects or excessive stress—and to guide optimization and problem solving.



### 2 INTERPRETATION CHART — VISCOSITY CURVE REGIONS



### 3 PROCESS OPTIMIZATION DECISIONS



**SELECT INJECTION SPEED / FILL TIME**  
Pick a speed that targets the stable window for your resin and part.



**SET TRANSFER POSITION**  
Switch from fill to pack at a consistent position for each shot.



**PLAN PACK / HOLD STUDY**  
Determine pack pressure and time needed to eliminate sink without overpacking.



**CONFIRM GATE SEAL STUDY**  
Verify gate freeze time vs. required hold to prevent flash.



**REDUCE LOT-TO-LOT SENSITIVITY**  
Operate in the center of the window. Tighten controls on speed and temperature.



**IMPROVE PROCESS TRANSFER BETWEEN PRESSES**  
Match fill time, pressures, and transfer position—not just speed settings.

### 4 COLD RUNNER ACTIONS



**REVIEW GATE AND RUNNER RESTRICTION**  
Ensure total flow path supports the target viscosity window.



**CONSIDER GATE SIZE / LOCATION CHANGE**  
Optimize gate to balance fill efficiency and gate freeze.



**MANAGE RUNNER COOLING**  
Control coolant temperature and flow for stable viscosity.



**CHECK COLD SLUG CONTROL**  
Ensure cold slug well size and purge frequency are adequate.



**WATCH GATE BLUSH OR HESITATION**  
Indicates low temperature, low speed, or restriction.

### 5 HOT RUNNER ACTIONS



**TUNE MANIFOLD / TIP TEMPERATURES**  
Set to keep resin in target window—neither too cool nor too hot.



**INSPECT TIPS FOR BLOCKAGE OR WEAR**  
Obstructions increase shear and cause inconsistent flow.



**MANAGE RESIDENCE TIME**  
Minimize heat soak; adjust temperatures and cycle time.



**REDUCE DROOL / STRINGING**  
Optimize tip temperature, retract, and nozzle seal.



**EVALUATE VALVE GATE TIMING OR THERMAL GATE BEHAVIOR**  
Ensure consistent seal and open/close performance.

### 6 PREVENTIVE MEASURES



**RESIN SELECTION**  
Choose the right grade with appropriate MFR / viscosity.



**ADDITIVES / FLOW PROMOTERS**  
Use when needed, validate with testing.



**UNIFORM WALL THICKNESS**  
Avoid abrupt changes and heavy sections.



**PROPER VENTING**  
Provide adequate vents for air and gases.



**MACHINE MAINTENANCE**  
Screws, checks, nozzles, and heaters in good condition.



**CLAMP-FORCE VERIFICATION**  
Confirm adequate clamp for every tool and material.



**AUXILIARY TEMP CONTROL**  
Maintain stable mold and support systems (chillers, dryers).

### 7 CORRECTIVE ACTIONS IN PRODUCTION

Use viscosity curve knowledge to guide safe parameter changes.



**SHORT SHOT**  
BEFORE → AFTER  
Increase speed within window, raise melt/mold temp, check venting. Do not only add pack.



**FLASH**  
BEFORE → AFTER  
Reduce speed, verify clamp force, check transfer position and pack time.



**BURN MARKS**  
BEFORE → AFTER  
Reduce speed, lower melt temp, increase venting, check tip temperature.



**FLOW HESITATION**  
BEFORE → AFTER  
Increase speed (still in window), raise temp, check gate/runner restriction.



**GLOSS VARIATION**  
BEFORE → AFTER  
Adjust speed to center of window, stabilize temps, check mold temp balance.



**GATE VESTIGE**  
BEFORE → AFTER  
Increase hold time if needed, optimize gate size or freeze, adjust transfer position.



**THE VISCOSITY CURVE DOES NOT REPLACE ENGINEERING JUDGMENT.** Optics, insert molding, or highly cosmetic parts may require additional caution. Use part-specific testing, DOE, and domain expertise to confirm the best settings.



**GOAL: CHOOSE A STABLE SPEED WINDOW, THEN BUILD THE REST OF THE PROCESS AROUND THAT FOUNDATION.**

Figure 8. Using the viscosity curve result to guide optimization and corrective action.

## 12.1 What a good curve looks like

- A steep early drop: the resin is shear-thinning and low-speed fill is unstable or pressure-intensive.
- A broad middle plateau: the preferred processing window; speed changes have less impact on apparent viscosity.
- A high-speed caution region: pressure may drop or behave erratically due to shear thinning, but risk increases for shear heat, burn, jetting, flash, material damage, and wear.
- Repeat points at the selected setting: pressure, fill time, cushion, part appearance, and weight should be stable.

## 12.2 Red flags in the graph

- No clear trend: data collection error, pressure limiting, changing short-shot target, check-ring leak, unstable temperature, or bad pressure data.
- Sudden pressure drop at one point: flashed mold, short-shot target changed, check-ring slip, air trap vented differently, or gate/runner temperature changed.
- Pressure increases at higher speed: venting limitation, high shear heating causing degradation/gas, gate restriction, pressure limit, clamp issue, or melt compressibility effect.
- Curve shifts between repeats: material moisture/lot, barrel soak, hot-runner drift, screw recovery instability, robot cycle changes, or mold temperature drift.
- Good curve but bad parts: selected speed may be rheologically stable but unacceptable for cosmetics, orientation, weld strength, dimensional stability, gate vestige, or residual stress.

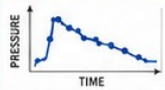


## 13. Diagnostic Techniques and Verification

# Viscosity Curves in Injection Molding

Advanced Diagnostics, Simulation, Case Examples, and Emerging Technology

### 1 DIAGNOSTIC AND VERIFICATION TOOLS



#### Pressure traces

- Fill, pack, and hold signatures
- Compare across lots and tools



#### Short-shot visual inspection

- Flow front pattern
- Weld lines, hesitation, knit lines



#### Gate freeze checks

- Determine freeze time
- Validate fill time vs. gate seal



#### Microscope / part surface inspection

- Flow marks, splay, burn, die lines
- Surface replication and gloss



#### CT scanning or X-ray

- Detect voids, knit weakness, sink, delamination
- Internal defect mapping



#### Capillary rheometry / material lab testing

- Generate viscosity vs. shear rate at multiple temperatures
- pVT (density) and moisture data



A viscosity curve is a **SETUP DOE**.

Once production starts, you do not "see" the curve on the part.

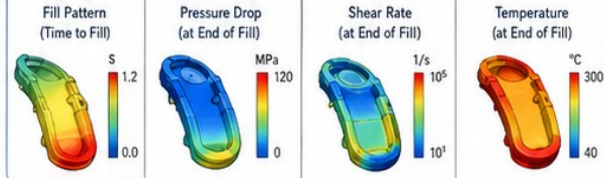
Instead, you infer its influence from:

- ✓ Process documents and settings
- ✓ Fill time targets and consistency
- ✓ Stable pressure signatures across shots

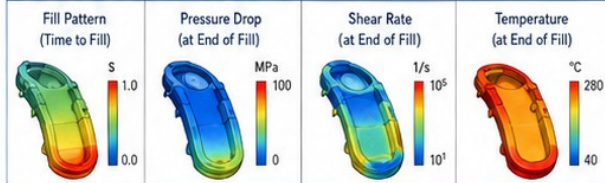
### 2 SIMULATION TOOLS

CAE helps predict how viscosity drives fill, pressure, shear, and temperature. Simulation quality depends on good viscosity and pVT data.

#### Moldflow-Style Analysis (Generic Example)



#### Moldex3D-Style Analysis (Generic Example)



Use simulation early to evaluate gate location, wall thickness, packing needs, and to compare resin or process options.

### 3 EXAMPLE APPLICATION CASES

#### A. MEDICAL THIN-WALL PART



**Problem:** Short shots and inconsistent fill in long, thin flow paths.

**Viscosity curve study revealed:** Higher viscosity at low shear than expected; strong shear-thinning knee causing hesitation.

**Production improvement:** Increased melt temperature and adjusted speed profile. Achieved complete fill with lower pressure and improved part-to-part consistency.

#### B. AUTOMOTIVE HOUSING



**Problem:** High pack pressure needed; warpage and sink at thick bosses.

**Viscosity curve study revealed:** Viscosity remained high at low shear (near-stall); limited shear-thinning benefit during packing.

**Production improvement:** Raised melt temperature and optimized pack/hold profile. Reduced pressure 20%, improved dimensional stability and surface appearance.

#### C. CONSUMER CLOSURE



**Problem:** Flash and high clamp force at high production speed.

**Viscosity curve study revealed:** Viscosity above target in high shear region; temperature sensitivity was strong.

**Production improvement:** Slightly higher melt temperature and faster fill in the stable window. Eliminated flash and increased output without quality loss.

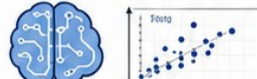
### 4 EMERGING TECHNOLOGY AND BEST PRACTICE

#### IoT MACHINE MONITORING



- Capture real-time pressure, speed, temperature, and alarms
- Trend process stability and detect drift
- Build run-to-run consistency dashboards

#### AI-ASSISTED PROCESS OPTIMIZATION



- Machine learning models learn from good runs
- Recommend optimal settings for quality and cycle time
- Detect anomalies early

#### CAVITY PRESSURE CORRELATION



- Validate pressure at the cavity, not just machine level
- Correlate pressure signature to part quality and fill time
- Improve packing and reduce variation

#### DIGITAL PROCESS DOCUMENTATION



- Store DOE results, viscosity curves, pVT data, and settings
- Link material lots, machine, mold, and environment
- Enable traceability and knowledge capture



#### STANDARDS & REFERENCES:

- ASTM D3835 / ISO 11443 – Capillary rheometry for polymer melts
- Best-practice guidance from scientific molding training, polymer suppliers, and industry organizations (e.g., SPE, Polymer Processing Society, and regional plastics associations)

#### GOOD PRACTICES:

- ✓ Use reliable data and controlled conditions.
- ✓ Repeat studies and confirm with production trials.
- ✓ Document assumptions and decisions.
- ✓ Share learnings across teams.

#### EXPERIENCED MOLDER ADVICE



**"Keep the study simple."**

Use a focused DOE and the few variables that really move viscosity.



**"Stabilize temperatures first."**

Mold, melt, and material temp stability matter more than chasing extreme settings.



**"Do not chase data from a bad machine."**

Fix the basics—maintenance, calibration, and consistency before trusting the results.



**"Document the chosen window."**

Write it down, share it, and protect it. Good processes are repeatable.

Figure 9. Advanced diagnostics, simulation, generalized case examples, and emerging technology.

A viscosity curve is not visually present on a molded part after production is running. You cannot look at a part and say with certainty that a viscosity curve was done. You infer it from documentation, fill-time control, stable pressure traces, consistent part weight, and a process that uses a rational speed window. Part defects can suggest that the chosen speed window is poor, but they do not prove whether the study was performed.

| Technique                            | What it checks                                                                                                      | Use                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Process documentation audit          | Check setup sheet, DOE worksheet, curve plot, selected fill time, transfer study, pack/hold study, gate seal study. | Best way to know whether the study was performed.                                                               |
| Pressure trace review                | Compare fill, transfer, pack, and hold signatures across shots/lots.                                                | Detects viscosity shifts, pressure limiting, check-ring issues, cavity restriction, and transfer inconsistency. |
| Short-shot visual inspection         | Observe fill pattern, flow front, weld locations, hesitation, race tracking, end-of-fill locations.                 | Directly supports curve interpretation.                                                                         |
| Microscopy/surface inspection        | Evaluate splay, burns, flow lines, gloss, knit lines, jetting, gate blush.                                          | Useful for defect mechanism, not proof of curve history.                                                        |
| Cavity pressure sensors              | Measure what the cavity sees, not only machine pressure.                                                            | High-value validation for medical, automotive, and critical parts.                                              |
| Part weight trend                    | Track shot-to-shot and lot-to-lot fill/pack consistency.                                                            | Good early warning of viscosity or transfer shifts.                                                             |
| CT/X-ray/ultrasound where applicable | Inspect internal voids, weak knit areas, thick-section sinks, inserts, and assemblies.                              | Useful non-destructive testing for critical components.                                                         |
| Material lab testing                 | Capillary rheometry, moisture, MFR, DSC, ash/filler content, FTIR, GPC where needed.                                | Confirms whether resin condition changed.                                                                       |

## 14. Moldflow, Moldex3D, Rheometry, and Alternative Methods

### 14.1 Simulation

Moldflow, Moldex3D, and similar CAE tools do not replace a press-side viscosity curve, but they can predict fill pattern, pressure drop, shear rate, temperature distribution, air traps, weld lines, gate options, clamp force, cooling effects, and whether a proposed design is likely to have a usable processing window. Simulation accuracy depends heavily on material characterization: viscosity versus shear rate and temperature, pvT behavior, specific heat, thermal conductivity, shrinkage data, and validation against real molding when possible.

### 14.2 Capillary rheometry and lab data

Capillary rheometry measures polymer melt viscosity over a range of temperatures and shear rates closer to processing conditions than a simple melt-flow-rate test. Standards such as ASTM D3835 and ISO 11443 describe methods for measuring rheological properties of polymer melts at relevant temperatures and shear rates. Use lab data when selecting materials, validating lot-to-lot changes, investigating unexplained viscosity shifts, or building high-confidence simulation material cards.

### 14.3 Other ways to get equivalent or better insight

- In-mold rheology curve with cavity pressure rather than machine pressure, when sensors are available.

- Instrumented nozzle or injection molding rheometer for material characterization.
- Mold-filling simulation using high-quality, recent material data and validated process conditions.
- Structured fill-time evaluation combined with cavity balance and pressure-loss study.
- Viscosity monitoring through peak pressure, fill integral, cavity pressure, and part weight trends in production.
- AI/analytics systems that learn normal pressure/fill signatures and flag drift - useful only if trained on good process data.

## 15. Preventive Measures

| Preventive area              | What to do                                                                                                         | Actionable recommendation                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Material selection           | Choose grade with appropriate flow, toughness, thermal stability, moisture tolerance, and regulatory requirements. | Ask resin supplier for processing guide, rheology data, drying requirements, and hot-runner limits. |
| Design for manufacturability | Uniform wall, proper flow length, radii, balanced thickness, ventable end-of-fill locations.                       | Do not expect speed to solve a bad flow path.                                                       |
| Gate and runner design       | Correct gate size, location, runner balance, cold slug wells, manifold/tip selection.                              | Design to keep the selected speed window practical.                                                 |
| Temperature discipline       | Control melt, mold, dryer, hot-runner zones, and cooling water.                                                    | Temperature drift is one of the most common false-curve causes.                                     |
| Machine maintenance          | Maintain screw, barrel, check ring, nozzle, heaters, sensors, hydraulics/electric drive, clamp.                    | A worn machine creates false data and poor transfer.                                                |
| Documentation                | Save the curve, raw data, chosen window, rejected points, defects, and assumptions.                                | Future processors need the rationale, not just the final setting.                                   |
| Training                     | Teach why the test is run, not just which buttons to push.                                                         | Processors who understand cause/effect are harder to replace and less likely to chase symptoms.     |

## 16. Corrective Actions

### 16.1 In-process adjustments

| Symptom    | Curve-guided action                                                                                                                                                                        | Do not forget                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Short shot | If speed is below stable window, increase speed within window; verify melt/mold temperature, venting, gate/runner restriction, pressure limit, and material dryness. Do not only add pack. | Adding pack cannot reliably fill a part that was not properly filled during first stage. |
| Flash      | Reduce speed if in high-speed risk region; verify clamp force, transfer position, pack pressure/time, mold parting line, vents, and material viscosity shift.                              | More clamp may hide but not solve overpacking or wrong transfer.                         |
| Burn marks | Reduce speed or move away from high-shear region; improve venting;                                                                                                                         | Burning can be trapped gas, material degradation, or both.                               |



|                      |                                                                                                                                |                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                      | lower overheated zones; check air traps and hot-runner tips.                                                                   |                                                                                  |
| Weld/knit weakness   | Move toward stable/faster fill if safe; raise melt/mold temp; improve venting/gate location; validate strength.                | Cosmetic improvement does not guarantee mechanical strength.                     |
| Splay/silver streaks | Check moisture, decompression, back pressure, screw recovery, hot-runner degradation, and excessive shear.                     | Do not call it a viscosity problem until moisture and degradation are ruled out. |
| Gate blush/jetting   | Reduce excessive gate shear, alter speed profile, adjust gate geometry, increase melt/mold temp, or use multi-stage injection. | Small gates can make speed changes look like material changes.                   |
| Gloss variation      | Center speed in stable window; stabilize mold temperature; check flow hesitation and wall thickness.                           | Gloss is thermal and rheological, not speed-only.                                |
| Dimensional drift    | Verify selected fill time, transfer, pack/hold, gate seal, cooling, material lot, and cavity pressure.                         | Viscosity curve is the starting foundation, not the full dimensional process.    |

## 17. Troubleshooting Flowchart and SOP

The following shop-floor logic should be printed and used before running the study. The point is to stop bad data before it becomes a bad process standard.

9. Confirm tool, cavity, gate, material, lot, and study objective.
10. Confirm mold and machine are mechanically healthy.
11. Confirm dryer, material handling, and moisture control.
12. Stabilize melt, mold, and hot-runner temperatures.
13. Disable pack/hold and establish repeatable short-shot target.
14. Confirm vents and clamp force before high-speed points.
15. Collect data at each speed only after the shot is stable.
16. Reject any point that is pressure-limited, flashes, changes fill target, or shows obvious machine malfunction.
17. Select speed window in the stable region and validate repeatability.
18. Document chosen fill speed/fill time and proceed to transfer, pack/hold, gate seal, cooling, and process window studies.

## 18. Case Studies and Field Accounts

The following are generalized training cases synthesized from common scientific-molding practice. They are not presented as named-company interviews. They should be used to teach diagnostic thinking, not as proof that a specific manufacturer experienced the exact event.

| Case                        | Problem                                                             | What the curve revealed                                                                                       | Resolution                                                                                                       |
|-----------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Medical thin-wall component | Long thin flow path; short shots at low speed; pressure near limit. | Curve showed steep viscosity drop from low to mid speed and a small stable region before cosmetic shear marks | Selected mid/high speed within stable window, increased verified melt temperature slightly, validated gate seal, |



|                         |                                                                                             |                                                                                                                                       |                                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                         |                                                                                             | appeared.                                                                                                                             | improved fill consistency.                                                                                       |
| Automotive housing      | Sink/warp at bosses and high pack demand; fill looked complete but pressure trace unstable. | Low-speed fill caused high apparent viscosity and late transfer inconsistency. Curve identified a stable fill time before pack study. | Set fill time, moved transfer, then optimized pack/hold. Reduced pressure variation and improved dimensions.     |
| Consumer closure - PP   | Flash at high-speed production target; clamp near limit.                                    | High-speed point was outside robust window; stable window existed at slightly lower speed with acceptable cycle impact.               | Reduced fill speed into stable window, adjusted transfer and pack. Flash reduced without increasing clamp force. |
| Hot-runner cap mold     | Cavity imbalance changed after lunch breaks and startups.                                   | Curve repeated poorly until manifold/tip temperatures and residence time were stabilized. One tip zone was drifting.                  | Repaired zone control, documented startup purge/soak rules, re-ran curve, then balanced cavities.                |
| Cold-runner ABS housing | Flow hesitation and gloss shift near gate; pressure high at all speeds.                     | Sprue bushing and small gate added restriction; speed alone could not create a broad stable window.                                   | Gate/runner redesign and mold-temperature control produced a more usable curve.                                  |

## 18.1 Technician and engineer field advice

- Keep the study simple. If more than one major variable changes, the curve becomes opinion, not data.
- Stabilize temperatures first. Mold, melt, material, and hot-runner stability matter more than perfect-looking spreadsheets.
- Do not chase data from a bad machine. Fix check rings, nozzles, pressure limits, and recovery instability before trusting a curve.
- Short shots must be comparable. Changing the fill target changes the pressure path and invalidates the point.
- The selected window must make acceptable parts. A stable curve point that creates unacceptable cosmetics, stress, or mechanical weakness is not the right point.
- Write down the why. The next processor needs to know why the speed was chosen and what limits were observed.

## 19. Industry Best Practices, Standards, and OEM Guidance

Several industry sources align on the core concept: viscosity is resistance to flow; thermoplastic melts commonly shear-thin; in-mold rheology/relative viscosity studies use pressure and fill time to choose a robust fill speed; and robust scientific molding relies on documenting plastic variables so processes can be transferred and repeated. Standards such as ASTM D3835 and ISO 11443 apply to capillary rheometry rather than shop-floor in-mold curves, but they define accepted laboratory methods for measuring melt rheological properties across temperatures and shear rates relevant to processing.

- Use press-side viscosity curves to establish the fill-speed/fill-time foundation of a process.
- Use laboratory rheology and material characterization for material selection, simulation, supplier disputes, lot investigation, and high-risk programs.
- Use simulation early for gate location, flow balance, pressure drop, air traps, shear heating, and cooling risk, but validate with actual mold data.

- Use supplier processing guides for drying, melt temperature, mold temperature, residence time, and safety constraints.
- Use machine OEM guidance for pressure limits, intensification ratio, screw/barrel/check-ring maintenance, clamp safety, and controller-specific data collection.
- Use hot-runner supplier guidance for manifold/tip zones, valve-gate timing, residence time, startup/shutdown, leakage, drool, and material-specific temperature limits.

## 20. Emerging Technologies and Research

- Cavity pressure and cavity temperature sensing provide better process visibility than machine pressure alone, especially in multi-cavity molds.
- IoT-enabled monitoring can trend fill time, peak pressure, cushion, recovery, mold temperature, dryer conditions, and hot-runner zones to detect drift before defects escape.
- AI-assisted process optimization can recommend process windows only when trained on clean, well-labeled, physically meaningful data. Bad DOE data creates bad AI decisions.
- Material digital twins and improved rheology models aim to improve simulation accuracy by using better viscosity, pVT, pressure-dependent viscosity, viscous heating, and thermal data.
- Advanced rheometry and injection molding rheometers can better represent processing shear rates and non-isothermal effects than simple MFR tests.
- Nano-additives, recycled-content materials, and high-filler compounds increase the need for real rheology data because small formulation changes can strongly alter shear and thermal response.

## 21. Training Checklists, Data Sheets, and Common Pitfalls

### 21.1 Pre-run checklist

- Correct mold, cavity selection, gate identification, and flow path documented.
- Correct resin, lot, color, additive, and regrind policy documented.
- Dryer conditions verified and moisture checked if required.
- Barrel zones stabilized and actual melt temperature checked.
- Mold temperature supply/return/flow stable; actual mold steel checked if possible.
- Hot-runner zones stable and no alarms if hot runner mold.
- Pack/hold disabled for study; short-shot target defined.
- Machine pressure limit high enough for velocity control, within safe limits.
- Clamp force adequate; vents clean; mold protection set.
- Cushion repeatable; check-ring condition acceptable.
- Robot/sprue picker and cycle automation stable.
- Data sheet ready; raw data and observations recorded immediately.

### 21.2 Common pitfalls

| Pitfall                | Damage to the study                                          | Prevention                                                                       |
|------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------|
| Changing fill target   | Pressure path changes; points cannot be compared.            | Use consistent short-shot fill percentage/position.                              |
| Running full parts     | Pack, gate seal, and part compression contaminate fill data. | Use short shots with pack/hold off unless engineering approves a special method. |
| Pressure-limited shots | The machine did not achieve                                  | Raise limit if safe or reject the point.                                         |

|                            |                                                         |                                               |
|----------------------------|---------------------------------------------------------|-----------------------------------------------|
|                            | commanded velocity.                                     |                                               |
| Wet material               | Hydrolysis, splay, bubbles, and viscosity change.       | Dry and verify material before study.         |
| Hot-runner drift           | Curve shows temperature control, not material response. | Stabilize and record all zones.               |
| Bad check ring             | Cushion and pressure inconsistent.                      | Repair machine before DOE.                    |
| Inadequate venting         | Pressure appears high; burns at fast speeds.            | Clean or repair vents before collecting data. |
| Overinterpreting the curve | Processor ignores part quality and downstream studies.  | Use curve as fill foundation only.            |

## 22. References and Further Reading

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## Appendix A. Blank Viscosity Curve Data Sheet

| Point | Speed setting | Actual fill time | Peak pressure | Pressure at endpoint | Cushion | Short-shot target | Melt temp | Mold temp | Observations |
|-------|---------------|------------------|---------------|----------------------|---------|-------------------|-----------|-----------|--------------|
|       |               |                  |               |                      |         |                   |           |           |              |
|       |               |                  |               |                      |         |                   |           |           |              |
|       |               |                  |               |                      |         |                   |           |           |              |
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|       |               |                  |               |                      |         |                   |           |           |              |

## Appendix B. One-Page Training Script

Teach the viscosity curve in this order:

39. Plastic is not water. Most melts shear-thin. Faster flow usually lowers apparent viscosity, but too fast can damage the material or process.
40. The curve tells us where speed is stable. We want the flat, robust region, not necessarily the fastest setting.
41. Only speed changes during the study. Temperature, material, target fill, pack, cushion, and machine condition must stay stable.
42. Cold runners add sprue/runner pressure loss and cold-slug risk. Hot runners add manifold/tip temperature, residence time, and gate-control risk.
43. Use the curve to choose fill speed/fill time. Then run transfer, pack/hold, gate seal, cooling, and process window studies.
44. A curve is only as good as the data. Bad machine, wet resin, poor venting, or drifting hot-runner zones create false conclusions.